# US distillate floor and refinery measurands: separated definitions for modeling

Co-authored, 2026-11-07. Status tags: MEASURED (published series), DERIVED (computed from published series), UNMEASURED (no open record found). Asterisks mark thin volume: more asterisks = thinner.

## Purpose

Public petroleum statistics come from separate surveys with different dates, scopes, denominators and end years. Single words ("utilization", "inventory", "capacity") cover several distinct measurands. This doc separates them so a model can sequence through defined terms instead of averaging welded ones. An unmeasured term is a gap to mark, not a zero.

# 1. Welded terms, separated

## Utilization

| ID | Measurand | Numerator / denominator | Source | Status |
|---|---|---|---|---|
| U1 | National crude-unit utilization | Gross inputs to crude distillation ÷ operable capacity (includes idle units restartable within 90 days), US aggregate | EIA weekly | MEASURED (92.8%, wk ending 2026-10-23) |
| U2 | Running-plant utilization | Same ratio, plants in operation only (turnarounds and idle excluded) | none | UNMEASURED |
| U3 | Bottleneck-unit utilization | Load on the most constrained downstream unit (hydrotreater, coker, etc.) per plant | operators, not published | UNMEASURED (operator reports ~98%, anecdotal) |

U1 ≠ U2 ≠ U3. U1 is pulled down by idle capacity and turnarounds; U3 can be pinned while U1 reads 92%.

## Distillate inventory

| ID | Measurand | Definition | Status |
|---|---|---|---|
| I1 | Total distillate stocks | Refineries + bulk terminals + pipelines + NG plants | MEASURED (weekly 99.3 M bbl, wk ending 2026-10-30) |
| I2 | Tank stocks | Refineries + terminals + NG plants | MEASURED monthly (100.2 M bbl, end 2025) |
| I3 | Pipeline fill | I1 − I2 | DERIVED (~28–34 M bbl/yr since 2000; 28.1 end 2025) |

| ID | Measurand | Definition | Status |
|---|---|---|---|
| I4 | Usable stock | Stock above the operating floor (section 2) | DERIVED, range only |

Weekly and monthly totals do not match (end-2025: monthly 128.2 vs weekly ~118). Mixing them is a basis error.*

### Storage capacity

| ID | Measurand | Definition | Status |
|---|---|---|---|
| C1 | Shell capacity | Physical tank volume | MEASURED: refineries to 2010; terminals to Mar 2024 (report discontinued) |
| C2 | Working capacity | Shell minus tank bottoms minus top contingency | same coverage as C1 |
| C3 | Unworkable fraction | (C1 − C2) ÷ C1 | DERIVED; survives scale and survey changes, which makes it the common denominator |

## 2. Operating floor equation

The structure is fixed. Values get swapped as data arrives.

```
FLOOR(t) = Σ_i [ shell_i(t) × u_i(t) ]      tank bottoms + top space
         + linefill(t)                      pipeline fill
         + holdup(t)                        process-unit inventory
         + working_min(t)                   stock needed to keep loading

i ∈ {refineries, bulk terminals}
u_i = unworkable fraction (C3)
USABLE(t) = I1(t) − FLOOR(t)
DAYS(t)   = USABLE(t) ÷ product supplied (b/d)
```

| Term | Value | Basis | Status |
|---|---|---|---|
| refinery u | 8.5% (2003) → 10.5% (2010); 11.8% (2016)* | EIA Jan 1 series; 2016 = Sept storage report, mixed basis | MEASURED/DERIVED |
| refinery shell | 70.6 M bbl (2010); 70.8 in operation (2016) | EIA | MEASURED, stale |
| terminal u | 9.3% (2016) | 191.6 shell vs 173.7 working | DERIVED, stale |
| terminal shell | 191.6 M bbl in operation (2016) | EIA storage report | MEASURED, stale (Mar 2024 file not yet read) |
| linefill | ~28 M bbl | I1 − I2 | DERIVED |
| holdup | none | not collected | UNMEASURED |
| working_min | none | not collected | UNMEASURED |
| standby/idle minimums | 2.6 M refinery + 2.9 M terminal idle shell (2016) | EIA; idle = restartable within 90 days | MEASURED, stale |

Current rough read (wk ending 2026-10-30): static floor ≈ 37–54 M bbl**; usable ≈ 45–62 M bbl**; ≈ 11–15 days at 4.175 M b/d**. The holdup and working_min terms would lower this further.

## Trend findings

- Pipeline fill has held roughly flat since 2000. The margin is shrinking from the top: tank stocks fell from 135.5 (2016) to 100.2 M bbl (2025).
- Refined-products pipeline miles: 62,391 (2004) → 65,060 (2025), +4%. Crude: 49,264 → 82,216, +67%. The new pipe raises the crude floor, not the diesel floor.
- ULSD switch (2006–2007): 15 ppm refinery tankage went from 3.6 to 37.3 M bbl by conversion, not addition. Grade segregation raises unworkable volume per barrel stored.
- The unworkable fraction rose 2003–2010 (+1.2 M bbl at refineries). It can't be split into bottoms vs top contingency from the data.**

- Old NPC floors (1979: distillate 125 M bbl against 365 M shell) were set on a larger system and can't be laid directly on current stocks.

## 3. Supply-side dependency tree (refining capacity)

```
REFINERY CAPACITY (new build or upgrade)
├─ Catalysts
|   └─ FCC catalyst: La/Ce ~2–3% by weight; ~12 t/day at a large refinery
|       └─ rare earth refining: China leading refiner, 19 of 20 strategic
minerals, ~70% avg share
|       └─ catalyst qualification 6–18 months (no fast swap)
├─ Heavy forgings (reactor vessels, hydrotreater shells)
|   └─ very heavy presses: Japan Steel Works, China (CFHI, Erzhong, Shanghai
Electric),
|      Doosan (KR), Le Creusot (FR), Russia (sanctioned); India building
|   └─ competing with nuclear buildout; JSW lead times 36 mo best, 5+ yr
practice
|   └─ refinery vs nuclear queue collision: UNMEASURED**
├─ Rotating equipment: compressors, turbines (Germany: Siemens, MAN), motors
|   └─ magnets: Nd, Dy, Tb (same rare earth processing bottleneck)
└─ Controls and electrical: chips, transformers (long lead times)
```

Case: Brownsville / America First Refining (~160,000 b/d). Permit issued 2021; three construction-start extensions; ownership Jupiter → Element Fuels → AFR; groundbreaking targeted Q2 2026, production 2028–29. Actual groundbreaking not confirmed. Parts supply is not named as a delay factor in coverage.

## 4. Demand-side stress

- Summer 2026 utilization (U1) highest since 1998. Capacity is down 4% since 2019, so output only matched 2019.
- Runs were driven by distillate exports into tight global supply. Stocks did not build before winter.
- Deferred turnarounds are likely to land in the peak distillate demand season. The exact deferral is UNMEASURED.

## 4b. Watch item: DEF branch and the farm collision

DEF is a second dependency chain that gates diesel transport independent of diesel supply. SCR-equipped engines (heavy trucks, 3/4- and 1-ton pickups, tow trucks, rail maintenance, road crews, construction, Tier 4 farm equipment) derate without it.

```
DEF (32.5% urea / 67.5% deionized water)
└─ urea
    └─ ammonia
        └─ natural gas (feedstock + energy)
    └─ Middle East production + shipping (2026 conflict disruption)
└─ competes with: urea fertilizer (spring planting demand)

FARM NODE (three draws, same season, shared upstream)
├─ diesel (field work)
├─ DEF (Tier 4 tractors)
└─ urea fertilizer
    └─ all three trace back toward natural gas and constrained supply
```

| Item | Value | Basis | Status |
|---|---|---|---|
| US bulk DEF move, early Apr 2026 | +15–20¢/gal | Argus via Mansfield Energy | MEASURED |
| Pilot most common retail DEF, spring 2026 | $4.299–4.599/gal | FreightWaves | MEASURED, dated |
| Upper Midwest pump DEF, 2026-11-07 | $4.79/gal | single station, operator report | OBSERVED, N=1* |
| US diesel retail, wk of 2026-10-26 | $6.35/gal (EIA); record $6.53 on 2026-09-22 (AAA) | EIA / AAA | MEASURED |
| DEF dose | ~2–3% of diesel burned | engine practice | approximate* |
| Uncompensated DEF cost per diesel gallon | ~10–15¢ | DERIVED from dose × $4.79 | DERIVED* |
| Fuel surcharge coverage of DEF | not in DOE diesel index; contract coverage not surveyed | — | UNMEASURED** |

| Item | Value | Basis | Status |
|---|---|---|---|
| TA/Petro DEF average | posted per location, not aggregated | — | UNMEASURED |

Welded term: "fuel cost." Surcharge and reimbursement formulas index diesel gallons only. DEF is a separate operating input that falls on the operator: owner-operators, small fleets, maintenance and municipal crews.

### Indicators to watch (diesel crunch + DEF)

1. East Coast (PADD 1) distillate stocks: a region can reach its floor before the national total does.
2. New York Harbor diesel prompt vs later-month spread: steep backwardation signals near-term shortage.
3. Distillate export volumes staying high into winter.
4. Bulk and retail DEF price vs diesel price ratio (currently ~0.75 at $4.79 / $6.35).
5. Urea fertilizer price into spring planting.

Cascade path (PROPOSED, not measured): diesel floor tightening + deferred turnarounds + DEF/urea competition at the farm node → simultaneous cost and availability pressure on freight, maintenance crews and planting.

## 4c. Watch item: sulfur branch and the food-transport loop

US sulfur is recovered as a byproduct, not mined (Frasch mining ended in the US around 2000). It comes from refineries and gas processing plants that strip hydrogen sulfide. Supply therefore tracks refinery runs: "plentiful in Texas" holds only while refineries run hard.

```
SULFUR (byproduct)
└─ refinery + gas plant H2S recovery
    └─ output falls with run cuts and turnarounds
└─ sulfuric acid
    ├─ refinery alkylation units (loop back to refining)
    ├─ phosphate fertilizer (farm node)
    └─ copper / metal processing

FOOD-TRANSPORT LOOP (not a hierarchy)
refining → diesel, sulfur → farm inputs (diesel, DEF, urea, phosphate)
farm output → needs diesel + DEF to haul
└─ no haul = crop stays in the field
```

Welded term: "sulfur supply is fine." Treats a byproduct as an independent stock. The coupling is that sulfur, diesel and fertilizer inputs all drop under the same refinery constraint at the same time.

Competition node: refining, fertilizer and metals draw on the same sulfuric acid. The farm node draws on diesel, DEF, urea and phosphate together. Stress on refining reaches both production and haul. A full harvest without haul capacity is lost output, not a surplus.

Status: structure DERIVED; magnitudes UNMEASURED**. Open: US byproduct sulfur volume vs refinery runs over time; sulfuric acid allocation across sectors under tightness.

## 4d. Thresholds from natural experiments: observed vs censored

Neither past event reached a break point, because the response stepped in first. Each one dates the stress signal; neither measures the failure threshold. Today's East Coast stocks are below the 2022 level, but the price spread isn't.

| Case | Stock / supply at stress | Price or response signal | Break point |
|---|---|---|---|
| A. Australia AdBlue, Dec 2021 | ~5 weeks AdBlue on hand + ~2 weeks inbound; fuel itself 98 days | Taskforce, local urea deal, manufacturers allowed to coordinate | Censored: intervention at ~5–7 weeks |
| B. US East Coast diesel, Oct 2022 | PADD 1 25.4 M bbl (Oct 28); US 25 days supply | NYH over Gulf ULSD +17 to +64 ¢/gal weekly (EIA monthly avg +56 ¢) | Censored: price pulled supply in; no prolonged outage |

| Case | Stock / supply at stress | Price or response signal | Break point |
|---|---|---|---|
| Now, Oct 2026 | PADD 1 21.0 M bbl (Oct 16); US ~23.8 days (99.3 / 4.175) | NYH over Gulf +5 to +27 ¢/gal weekly, Sep–Oct | Not reached |

- **Additive gate (case A):** transport can stop on a 5-week additive stock while fuel sits at 98 days. Model DEF days-on-hand as its own gate, not as part of diesel.
- **Signal divergence (now):** PADD 1 stock is ~17% below Oct 2022, yet the NYH–Gulf spread is about a third of the 2022 peak. Possible readings, untested: more Colonial or import flow, lower regional demand, or the spread lags the stock. Branches kept open.
- **Days-of-supply basis:** both cases and now use stocks ÷ 4-week product supplied. Weekly totals include pipeline fill (section 2), so days above the floor are fewer than the headline figure.
- **Sulfur coupling:** USGS ties refinery sulfur directly to utilization. The coefficient (USGS monthly sulfur by PADD vs EIA monthly runs) is not run yet: status unmeasured.

For modeling, tag each threshold with one of three statuses: observed (a signal value with date and source), censored (the response fired first, so the break point is unknown), or unmeasured.

Sources for 4d: EIA PADD 1 weekly distillate stocks · EIA NYH ULSD spot · EIA Gulf Coast ULSD spot · ACCC AdBlue authorisation · Guardian Australia, AdBlue · MarketNews, Oct 2022 days of supply · Mansfield / EIA, NYH spread Oct 2022 · USGS MCS 2026 Sulfur

## 5. Data gaps (pickup points)

1. U2 and U3 utilization: not published.
2. Refinery storage capacity after 2010: not published.
3. Terminal storage after March 2024: report discontinued; Mar 2024 file not yet read.
4. Process-unit holdup after upgrades (hydrotreaters, renewable diesel units): never collected.
5. Working minimum at terminals: not collected.
6. Regional floors: one region can run out before the US total does.
7. Forge queue allocation between refinery and nuclear orders.

## Sources


- EIA weekly petroleum status; distillate ending stocks (MDISTUS1); stocks at refineries/terminals/NG plants (MDISXUS1)
- EIA refinery shell and working storage capacity series (1982–2010), incl. sulfur-grade splits
- EIA Working and Net Available Shell Storage Capacity, Sept 2016 (archived); report page (discontinued after Mar 2024)
- EIA Today in Energy: refinery utilization 2026; working vs shell storage
- PHMSA hazardous liquid annual report mileage, 2004–2025
- NPC/DOE 1979 Petroleum Storage & Transportation Vol. II; NPC 1998/2004 inventory reports
- World Nuclear Association heavy manufacturing; IEA critical minerals commentary; CRS rare earths brief
- Inside Climate News, KXAN, BIC Magazine on the Brownsville refinery